

Analysis of Discrimination Complaint Cases in Iowa



Context

Filing a discrimination complaint is not just paperwork.

Many victims wonder:

- "Is this worth the emotional energy?"
- "Do I actually have a winning case?"

Our Goal is to understand **key factors** of discrimination cases in Iowa, and develop a **predictive model** to make recommendations to complainants pursuing legal action.

Business Problem

- Is it possible to preemptively predict the outcome of a case?
- Which characteristics, demographics and types of discrimination cases predict the outcome of the case?
- How can we make recommendations to clients to pursue legal action in discrimination cases?

A good outcome is a model that correctly predicts a favorable outcome 75% of the time.

Data Overview

Iowa Government (Open Data) *Closed Discrimination Complaint Cases in Iowa*

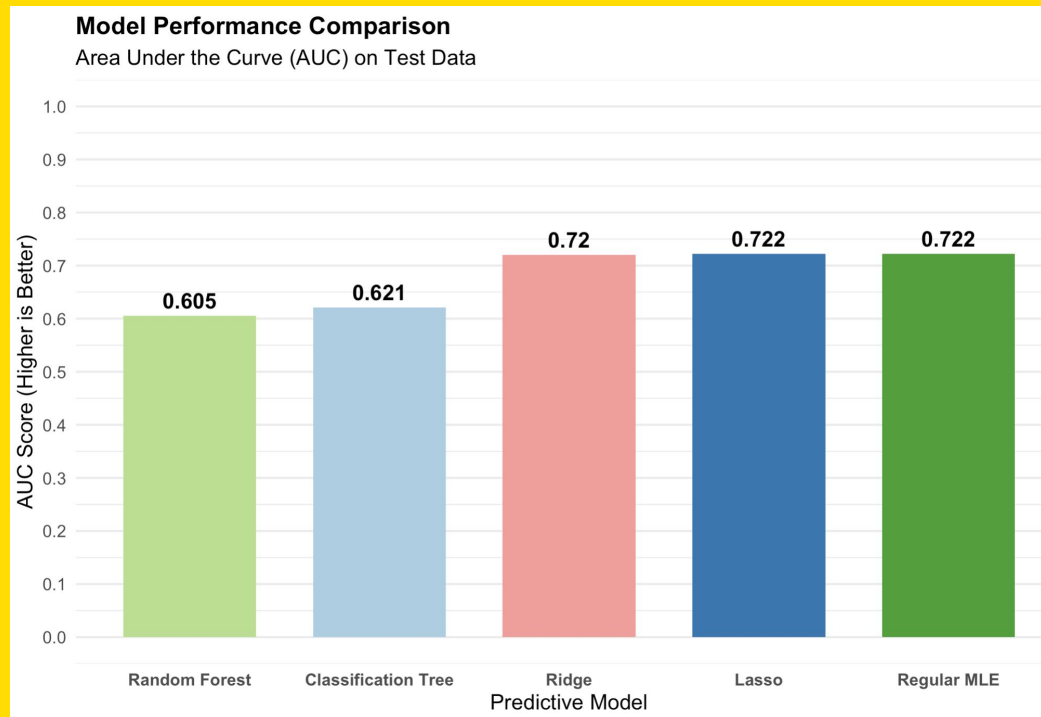
Time Period: 2008 to 2026 (Fiscal Year)

Predicting: Favorable Outcome vs. Unfavorable Outcome

- A **favorable** outcome is when the client received a tangible, positive outcome.
- An **unfavorable** outcome is any procedural or merits-based closure that did not result in a tangible win for the client.

Methods - Predictive Models

We compared several predictive models including a classification tree, random forest, and a logistic regression fit with MLE, lasso, and ridge penalization...



Methods - Predictive Models

Although the MLE and lasso had the same AUC, we chose a logistic regression model with **lasso penalty** as our final model because sensitivity was a priority. Lasso had a 3% better sensitivity.

Evaluation Metrics:

If we set $\pi^* = .162$ (threshold that we set for prediction), we are estimated to get a specificity of .693 and sensitivity of 0.601.

- That is, we will predict an unfavorable outcome 69% of the time when the outcome is actually unfavorable.
- Further, **we will predict a favorable outcome 60% of the time when the outcome is actually favorable.**

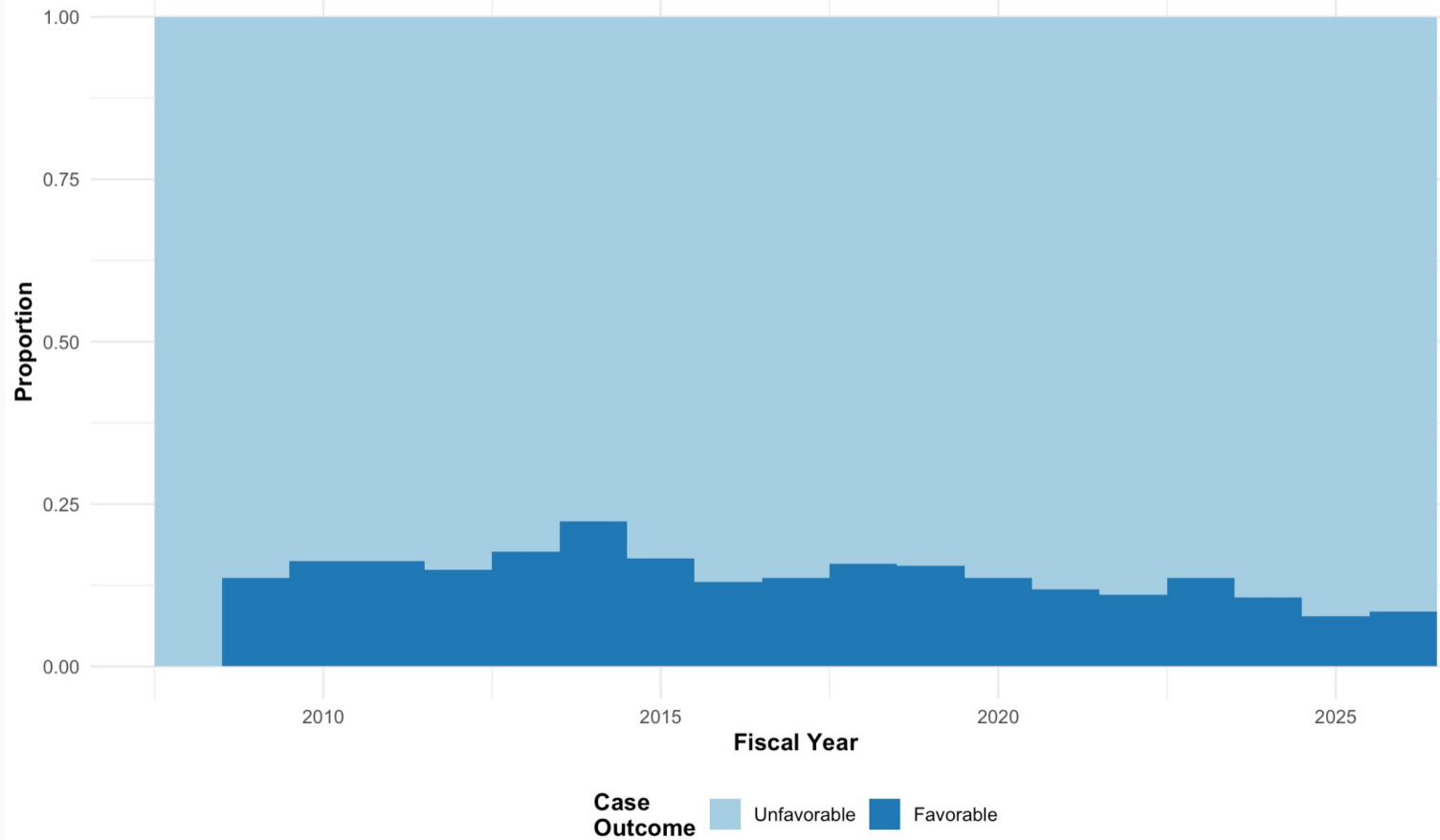
Methods - Descriptive Model

We supplemented our predictive model with a **descriptive** logistic regression model for easier interpretation. This allows us to provide **actionable advice to the clients**.

Our lasso predictive model deemed variables as the most important predictors of Outcome, so we built the model with these:

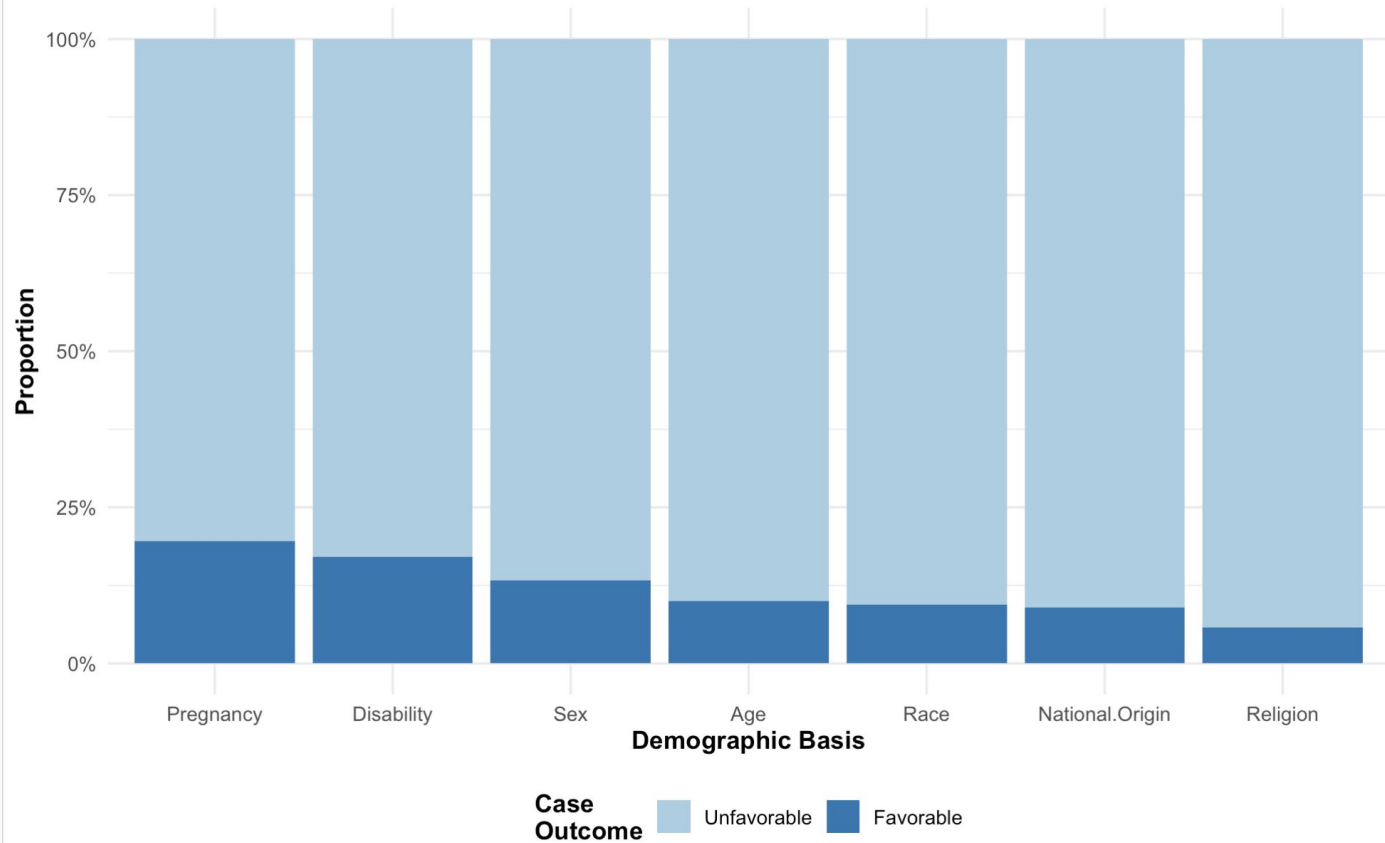
- Processor
- Housing
- Public.Accommodations
- Education
- Credit
- Race.Type
- Disability
- Age
- Sex.Type
- Pregnancy
- National.Origin.Type
- Marital.Status
- Religion.Type
- Creed
- Color
- Sexual.Orientation

Case Outcomes over Time (Fiscal Year)



Favorable Outcome Rates by Demographic Basis

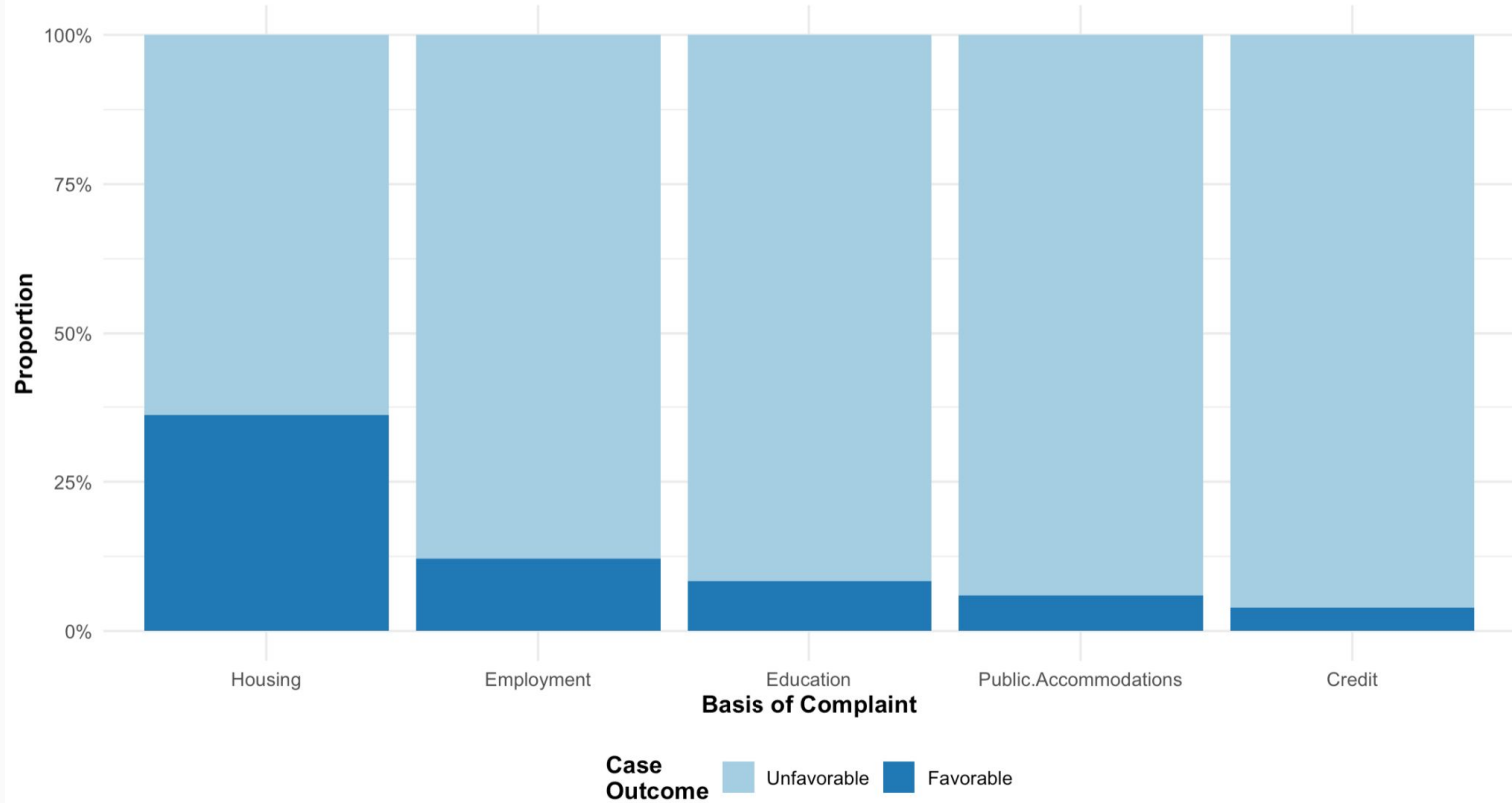
Proportion of Favorable vs. Unfavorable outcomes, sorted by Favorable rate



Data: Iowa Civil Rights Commission

Favorable Outcome Rates Vary by Complaint Basis

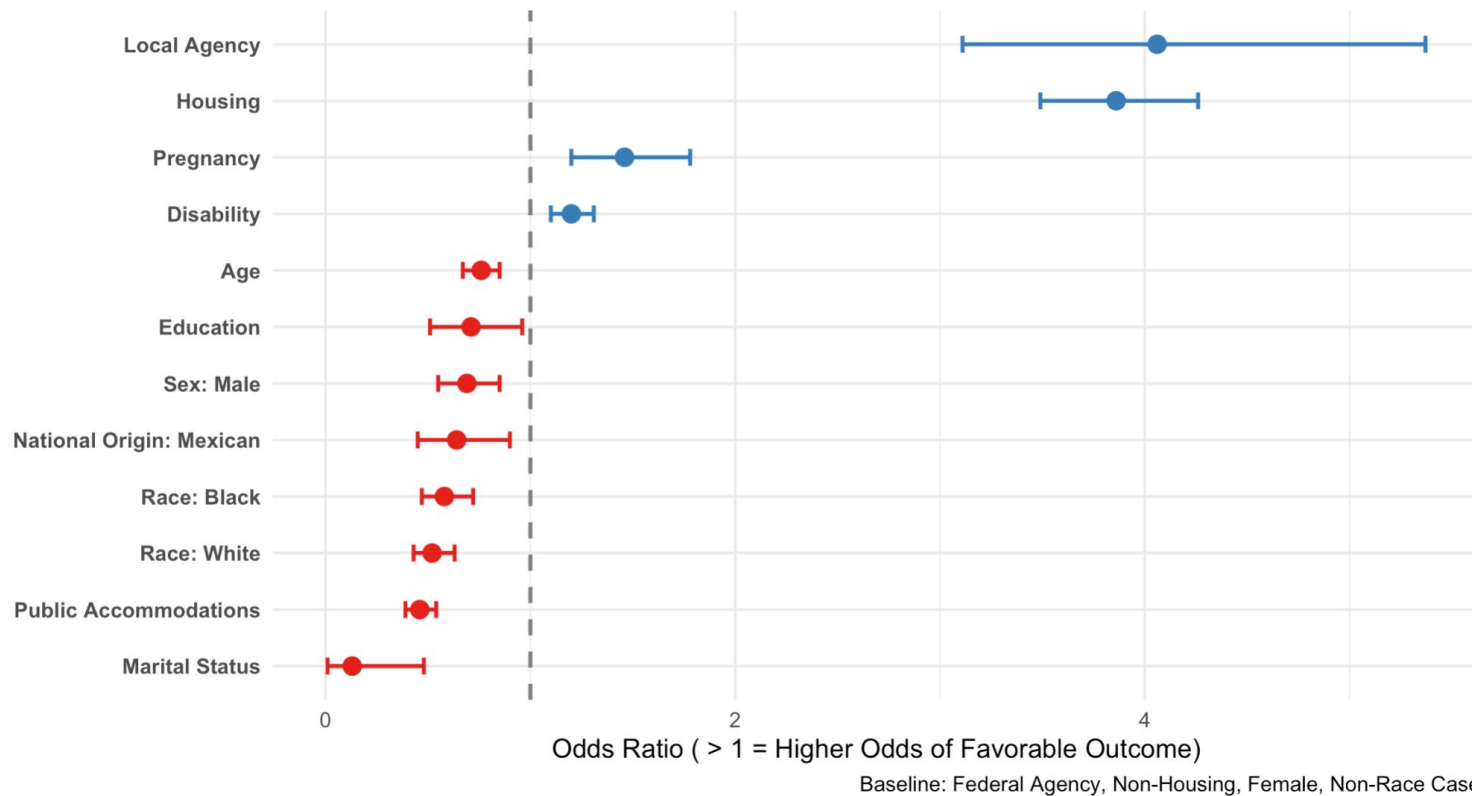
Proportion of Favorable vs. Unfavorable outcomes, sorted by Favorable rate



Data: Iowa Civil Rights Commission

What Drivers Increase Your Odds of Success?

Odds Ratios (95% Confidence Intervals) relative to the Baseline



The odds of a favorable outcome for complaints filed based on creed, color, credit and sexual orientation are not significantly different than not filing on those basis.

Interpretations – Increased Odds

Where you file your case matters

- All else held constant, **filing a claim locally has odds of achieving a positive outcome over 4 times greater than those filed at the EEOC** (federal level). We are 95% confident that the true change in odds is between a factor of 3.1 and 5.4.

Housing cases are successful

- All else held constant, **the odds of a positive outcome for claims filed for housing complaints are 3.9 times higher than non-housing claims**. We are 95% confident that the true change in odds is between a factor of 3.5 and 4.3.

Pregnancy cases are successful

- All else held constant, **the odds for a positive outcome for claims filed on the basis of the complainant's pregnancy are 46% more than non-pregnancy based claims**. We are 95% confident that the true value is between 20% and 78% increase.

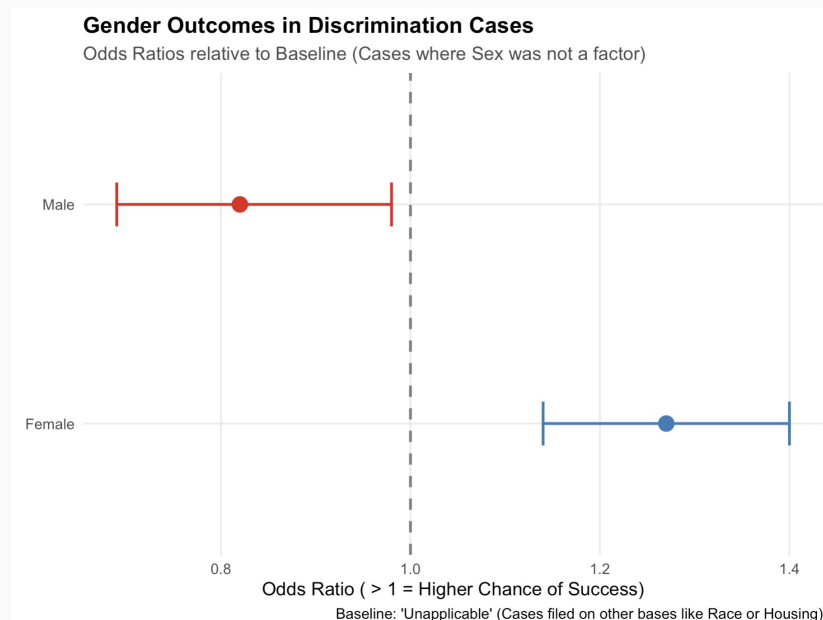
Disability cases are successful

- All else held constant, **the odds for a positive outcome for claims filed on the basis of the complainant's disability are 20% more than non-disability**. We are 95% confident that the true value is between 10% and 31% increase.

Interpretations – Increased Odds

Females are successful

- Female complainants have significantly higher odds of success than cases where Sex was not a factor. **We are 95% confident that being Female increases the odds of a favorable outcome by 14% to 40%.**
- We are 95% confident that being Male decreases the odds of a favorable outcome by roughly 2% to 31% compared to non-sex-based claims.



Interpretations – Decreased Odds

Marital status claims are not successful

- All else held constant, **the odds of a positive outcome for complaints on the basis of marital status are 88% less than non-marital status based claims.** We are 95% confident that the true value is between 52% and 99% less.

Public accommodations cases are not successful

- All else held constant, **the odds for a positive outcome for complaints associated with public accommodations (public services and buildings) are 54% less than non-public accommodation based claims.** We are 95% confident that the true value is between 47% and 60% less.

Education cases are not successful

- All else held constant, **the odds for a positive outcome for complaints associated with education are 29% less than non-education based claims.** We are 95% confident that the true value is between 4% and 49% less.

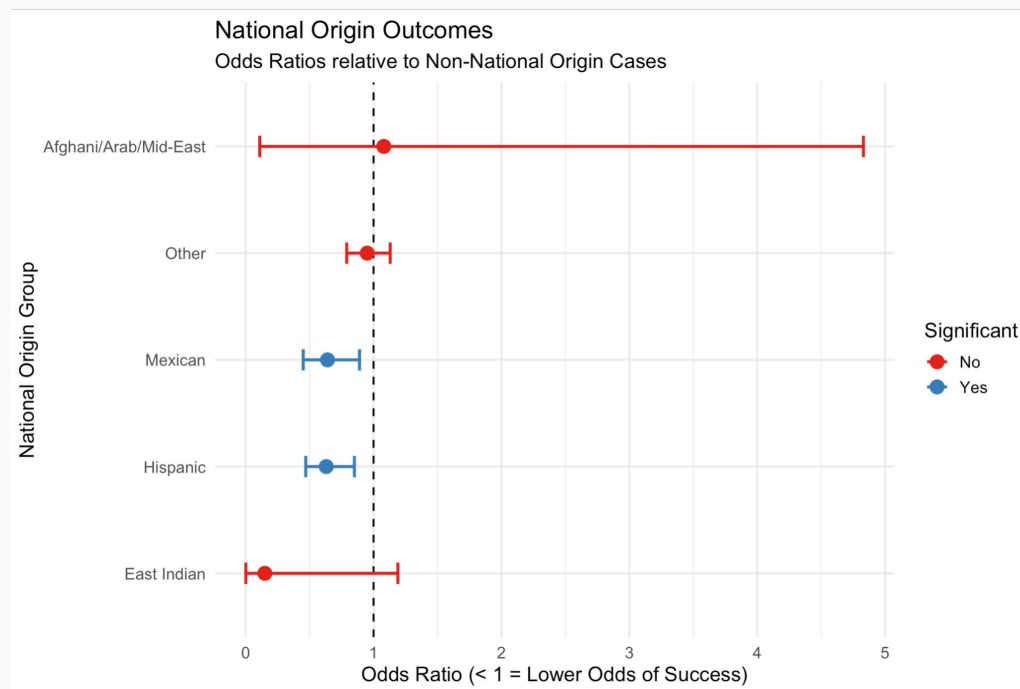
Age cases are not successful

- All else held constant, **the odds for a positive outcome for claims filed on the basis of the complainant's age are 25% less than non-age based claims.** We are 95% confident that the true value is between 15% and 33% less.

Interpretations – Decreased Odds

National Origin cases are not successful

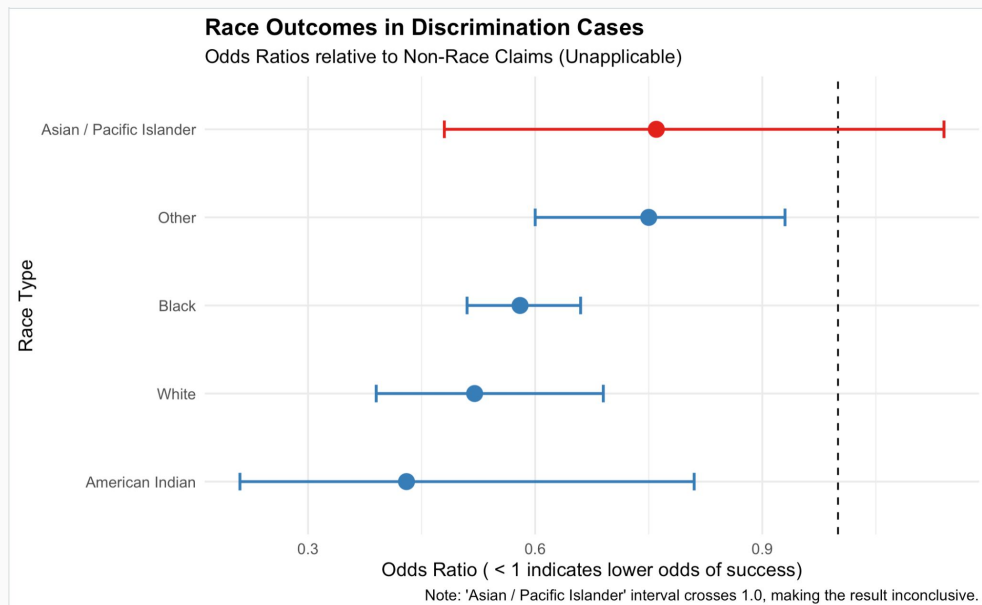
- All else held constant, the only two national origin types that have a statistically significant difference in odds of a positive outcome compared to not filing for **national origin based discrimination** are types **"Mexican"** and **"Hispanic"**. **The odds for these both are around 36% less than not filing for national origin based discrimination.** We are 95% confident that the true value is between 11% and 55% less.



Interpretations – Decreased Odds

Race-based discrimination cases are not successful

- All else held constant, claims based on American Indian, Black, White, or 'Other' races have significantly lower odds of a positive outcome compared to non-race claims. These groups face odds that are between **25% and 57% lower than non-race based claims.**
- The only race type that did not have a significant difference from non-race based claims was Asian/Pacific Islander.



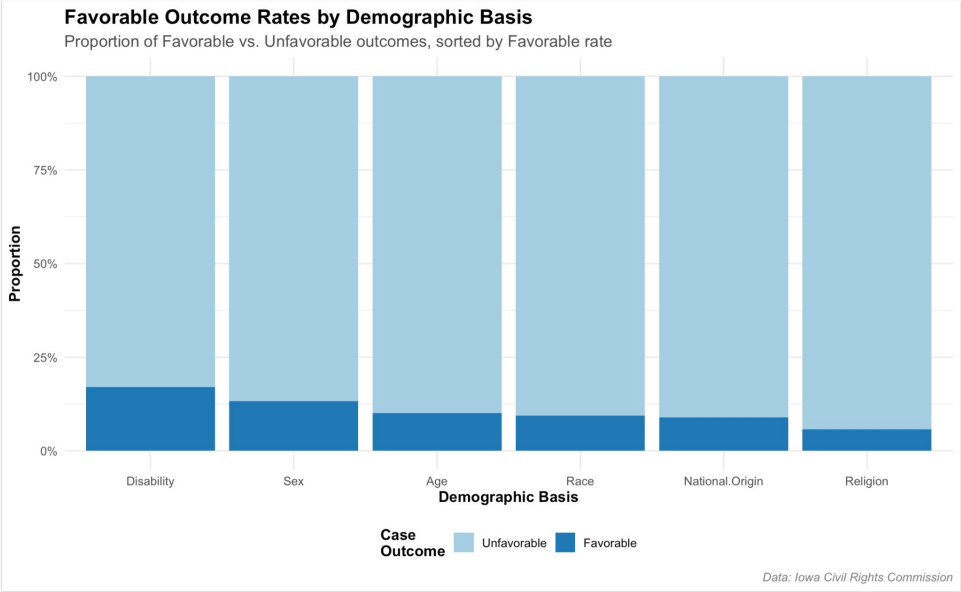
Results

What did we learn?

1. Filing location and discrimination type are the strongest predictors of case outcome.
2. Several demographics (especially race, national origin, religion, and age) are associated with a *decreased* multiplicative shift in odds of a favorable outcome.
3. Sex (filed by women), housing, pregnancy, and disability discrimination cases have a multiplicative *increase* in odds of obtaining a favorable outcome.

Key Insights:

- Where you file your claim matters – Local claims 4x odds (compared to federal level)
- Housing (3.9x), pregnancy (1.5x), and disability (1.2x) discrimination cases result in increased odds of favorable case outcomes
- Race (-36%), sex (-2% to 31% for males), age (-25%), education (-29%), marital status (-88%), and public accommodations (-54%) discrimination cases result in decreased odds of favorable case outcomes



Recommendations

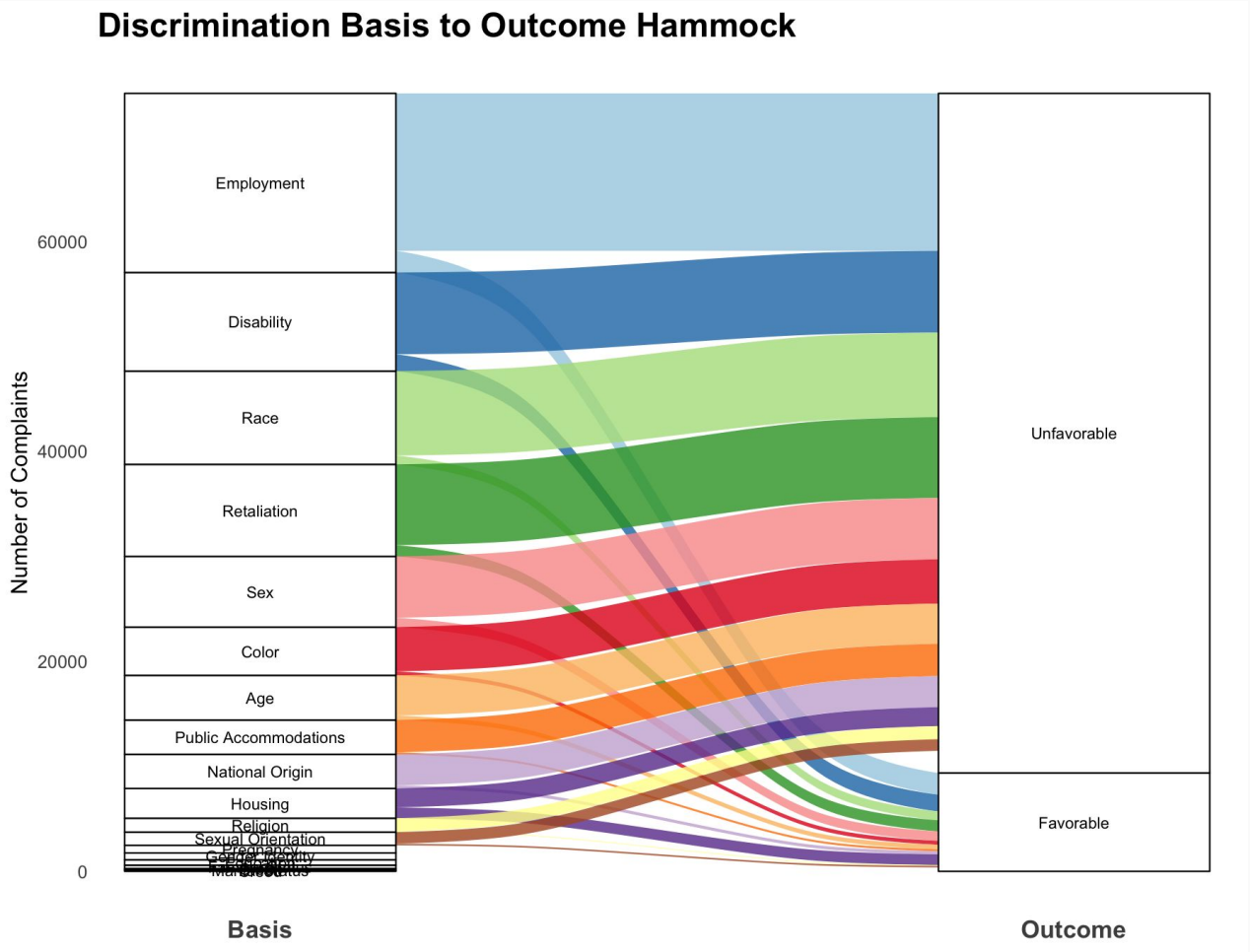
We recommend to file cases locally, and to pursue legal action for housing, disability, and pregnancy discrimination cases.

Increasing Odds (In order of priority)

- 1. Processing Type
- 2. Housing
- 3. Pregnancy
- 4. Disability

Decreasing Odds

- 5. Age
- 6. Education
- 7. Sex (for males)
- 8. National Origin
- 9. Race
- 10. Public Accommodations
- 11. Marital Status



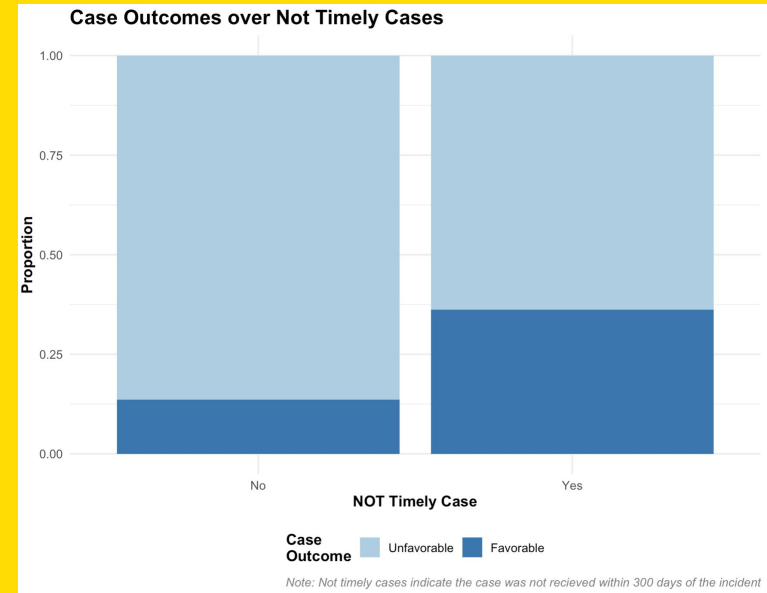
Limitations

Our goal for a good outcome was a model that correctly predicts a favorable outcome 75% of the time. **We only achieved this 60% of the time.**

A lot of variables had to be excluded in order to make it a **screening tool**.

There were issues in our model due to several variables based on *time*.

- "Not Timely" variable → **"Survivor's bias"**



Conclusions

Problem: To understand key factors of discrimination cases in Iowa, and develop a predictive model to make recommendations to complainants seeking to pursue legal action.

Method: We compared several predictive models including a classification tree, random forest, and a logistic regression fit with MLE, lasso, and ridge penalization and we chose lasso as our final model because sensitivity was a priority. To make interpretations, we fit a linear regression model using the variables deemed important by that lasso model.

Takeaways:

- Several discrimination cases such as **race, national origin, religion, and age are associated with a decrease in the odds of a favorable outcome.**
- Our model showed that we can use Iowa discrimination case to predict a future case outcome: **out of 100 positive outcomes, we will correctly predict 60.** We recommend to **file cases locally, and to pursue legal action for housing, disability, and pregnancy discrimination cases.**

Q&A

Any questions?
Ask away!